SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 Duration : 1 Hour Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ *(Name / Reg No) certify that this submission is my original work*
Thuviksha V / 1924260035
and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

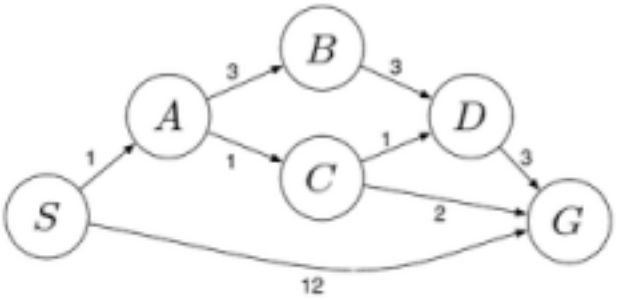
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



ANALYTICAL PROBLEM SOLVING — ANSWERS

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1. Water Jug Problem

Goal: obtain exactly 2 gallons in the 4-gallon jug.

Step	Action	4-gallon	3-gallon
1	Fill 3-gallon jug	0	3
2	Pour into 4-gallon jug	3	0
3	Fill 3-gallon jug again	3	3
4	Pour into 4-gallon jug until full	4	2
5	Empty 4-gallon jug	0	2
6	Pour remaining water into 4-gallon jug	2	0

Result: Exactly 2 gallons remain in the 4-gallon jug.

2. Mars Rover Intelligent Agent

i. Percepts: Camera images; rock/soil sensors; scientific instruments; temperature and navigation sensors. **ii.**

Environment: Partially observable, sequential, dynamic, continuous, and mainly single-agent; outcomes can be uncertain.

iii. Actions: Move and turn, capture images, collect and analyse samples, avoid obstacles, and transmit data to Earth.

iv. Performance: Quantity and quality of samples, accuracy of data, area explored, obstacle avoidance, efficient battery use, and successful data transmission.

v. Architecture: A model-based, goal-based agent is suitable because it maintains information about surroundings and plans actions to achieve exploration and sample-collection goals.

3. Eight-Queens Problem

Initial state: Empty 8 × 8 board. **State space:** Possible queen placements. **Actions:** Place a queen in a safe square, one column at a time. **Goal test:** Eight queens are placed with no two in the same row, column, or diagonal. **Path cost:** One per placement; total 8 placements.

Search strategy — Backtracking: Start at the first column and place a queen in a safe square. Continue column by column. If a column has no safe square, return to the previous column and move that queen. Repeat until all eight queens are safely placed.

ANALYTICAL PROBLEM SOLVING — ANSWERS (CONTINUED)

4. OLA Cab Booking Application

Agent type: Goal-based problem-solving agent. It treats the destination as the goal and helps the customer choose a suitable cab and route.

Pseudocode

BEGIN
1. Get customer's current location.
2. Get destination location.
3. Display cab types: Mini, Micro, Sedan, Shared, Prime, etc.
4. Get customer's cab choice.
5. Find available cabs.
6. Calculate estimated fare and travel time.
7. Display details to the customer.
8. If customer confirms, book the cab and display driver/trip details; otherwise cancel booking.
STOP

Result: The customer can choose a cab and book a ride to the destination.

5. Uniform Cost Search (UCS)

Algorithm: UCS finds a least-cost path by expanding the frontier node with the lowest cumulative cost.

BEGIN
1. Insert start node S into a priority queue with cost 0.
2. While the queue is not empty, remove the node with minimum cost.
3. If the node is G, return its path and total cost.
4. Otherwise, expand its neighbours and calculate each new cumulative cost.
5. Add or update a neighbour when the new path is cheaper.
6. Repeat until G is reached.
7. If the queue becomes empty, report “No path found”.
STOP

Result: UCS returns the least-cost path when all route costs are non-negative. The weighted graph is not visible in the supplied PDF, so the exact path and numerical cost cannot be calculated from this file.